

Solution: By defn. of invertible matrices, there exists a unique $n \times n$ matrix A^{-1} such that $A^{-1}A = AA^{-1} = I_n$.

• First let's verify whether $A^{-1}B$ is a solution of the equation $AX=B$.

$$A(A^{-1}B) = (AA^{-1})B \text{ (Theorem 2(a))} \\ = I_n B \text{ (By definition)} = B \text{ (Theorem 2(e))}$$

• Now, we shall prove the uniqueness of the solution $A^{-1}B$.
 $A^{-1}(AX) = A^{-1}B \Rightarrow (A^{-1}A)X = A^{-1}B \text{ (Theorem 2(a))} \Rightarrow I_n X = A^{-1}B$
 (By definition of inverse) $\Rightarrow X = A^{-1}B \text{ (Theorem 2(e))} \square$

12) Let A be an invertible $n \times n$ matrix, and let B be an $n \times p$ matrix. Explain why $A^{-1}B$ can be computed by row reduction:

If $[A \ B] \sim \dots \sim [I \ X]$, then $X = A^{-1}B$.
 If A is larger than 2×2 , then row reduction of $[A \ B]$ is much faster than computing both A^{-1} and $A^{-1}B$.

Solution: From the proof of Theorem 7, there exists elementary matrices $E_1, E_2, \dots, E_p$ s.t. $(E_p E_{p-1} \dots E_2 E_1)A = I$ and $(E_p E_{p-1} \dots E_2 E_1)B = A^{-1}B$.

$$I = A^{-1}A \Rightarrow B = I B \text{ (Theorem 2(e))} \\ \therefore (E_p E_{p-1} \dots E_2 E_1)B = (E_p E_{p-1} \dots E_2 E_1)I B \\ = (E_p E_{p-1} \dots E_2 E_1)I B \text{ (Theorem 2(a))} \\ = A^{-1}B \text{ (Theorem 7).}$$

$\therefore [A \ B]$ can be row reduced to $[I \ A^{-1}B] \square$

13) Suppose $AB=AC$, where B and C are $n \times p$ matrices and A is invertible. Show that $B=C$. Is this true, in general, when A is not invertible? (True) $n \times n$

Solution: By the defn. of invertible matrix, there exists a unique matrix A^{-1} such that $A^{-1}A = AA^{-1} = I_n$.

$$AB=AC \Rightarrow A^{-1}(AB) = A^{-1}(AC) \Rightarrow (A^{-1}A)B = (A^{-1}A)C \text{ (Theorem 2(a))} \\ \Rightarrow I_n B = I_n C \text{ (By defn.)} \Rightarrow B=C \text{ (Theorem 2(e))}$$

The condition that $AB=AC$ is not true in general, when A has not an inverse